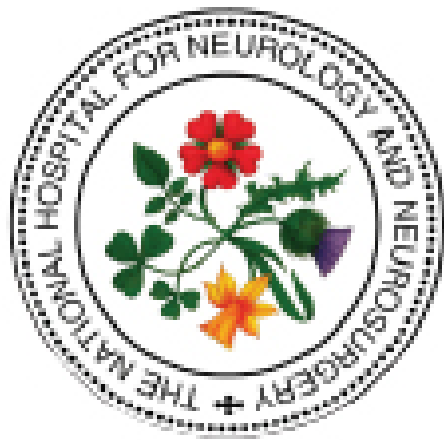




ACUTE INTRACEREBRAL HAEMORRHAGE (ICH)

ICH is a medical emergency; its rapid diagnosis, investigation and treatment can prevent further brain injury and improve outcome. Full supportive care should be considered for most patients for at least 24–48 hours, as prognostication can be difficult. Hyper-acute interventions such as reversing anticoagulation, BP control, neurosurgery and management in an acute stroke unit or, if required, a neuroscience critical care unit may improve survival.

This protocol outlines safe targets based on agreed guidelines, which may be adapted in individual clinical circumstances with the judgement / knowledge of the responsible senior clinician

DEFINITION

Intracerebral haemorrhage (ICH) is defined as acute blood extravasation into brain parenchyma from a ruptured cerebral blood vessel. It is the 2nd most common (15-30%), and deadliest form of stroke, associated with high mortality and morbidity.

ICH can be due to cerebral small vessel disease, cerebral amyloid angiopathy, result of macrovascular disorder (AVM, cavernoma, aneurysm, venous sinus thrombosis) or a neoplasm. ICH may be linked to hypertension or anticoagulation therapy. Delayed deterioration may occur due to re-bleeding, swelling, hydrocephalus, seizures.

CURRENT ICH GUIDELINES

<https://www.ahajournals.org/doi/suppl/10.1161/STR.0000000000000407>

IMAGING

1. Non-contrast brain CT study of choice in acute phase
2. CTA performed acutely to rule out a macrovascular cause, preferably within 2 days of non-contrast CT
3. CTA may also show predictors of expanding haematoma such as the 'spot sign' (foci of contrast extravasation within the haematoma)
4. MR / MRA recommended (once patient stabilised) to provide information about underlying cerebral small vessel disease
5. DSA if at high risk of underlying macrovascular cause

TREATMENT

Anticoagulation reversal: Risk of haematoma expansion, rapid deterioration, and poor outcome increases with anti-coagulation linked ICH - *anticoagulation should be stopped immediately and rapidly reversed.*

Intensive BP management: Acute BP control to achieve target of $\leq 140\text{mmHg}$ for at least 7 days based on agreed guidelines – *patient specific BP decisions also based on imaging.*

Surgery to remove clot: Expanding haematoma (which occurs very early) and vasogenic oedema can lead to an increase in ICP and fatal brain herniation syndromes - *craniotomy and clot removal may be performed for surgically accessible superficial bleeds.*

If left, brain will eventually re-absorb clot, however, the damage done by increased pressure over a long period may be irreversible.

